Experiment-7

Aim

Write a MATLAB program to transfer an analogue transfer function to a digital transfer function using Infinite Impulse transform.

Apparatus
Academic License MATLAB (Version 2025b)

Theory

1. Need for IIR Filters

Digital filters are widely used in signal processing to remove noise, shape signals, or extract useful information. Among them, **Infinite Impulse Response (IIR) filters** are important because their impulse response theoretically continues indefinitely.

The need for IIR filters arises due to the following reasons:

- **Efficient implementation:** IIR filters require fewer coefficients compared to FIR filters to achieve a similar frequency response.
- **Lower computational complexity:** They use feedback, which reduces the number of calculations.
- **Memory efficiency:** Since fewer filter coefficients are required, memory usage is reduced.
- **Ability to mimic analogue filters:** IIR filters can closely approximate analogue filter responses such as Butterworth, Chebyshev, and Elliptic filters.

Because of these advantages, IIR filters are commonly used in real-time digital signal processing systems.

2. Different Methods to Design IIR Filters

IIR filters are generally designed by transforming an **analog filter** into a **digital filter**. The most common design methods are:

1. Impulse Invariance Method

This method converts an analog filter into a digital filter by sampling the impulse response of the analog system. The digital impulse response is made identical to the sampled analog impulse response.

2. Bilinear Transformation Method

This method maps the analog frequency axis into the digital frequency axis using a nonlinear transformation. It avoids aliasing but introduces frequency warping.

3. Matched Z-Transform Method

In this technique, each pole and zero of the analog filter is mapped directly into the z-plane using exponential mapping.

4. Approximation-Based Methods

Analog filters such as:

- Butterworth filters
- Chebyshev filters
- Elliptic filters

are first designed and then converted into digital filters using transformation techniques.

3. Impulse Invariance Design Technique

The **Impulse Invariance Method** converts an analog transfer function into a digital transfer function by **sampling its impulse response at a fixed sampling period** .

If the impulse response of the analog system is:

then the digital impulse response is obtained as:

This ensures that the **time-domain characteristics of the analogue system are preserved**.

In practical implementations, software tools such as **MATLAB** provide the function:

impinvar

This function automatically converts the analogue filter coefficients into digital filter coefficients and generates the digital transfer function using the **tf** command.

Thus, the digital filter behaves similarly to the original analogue filter when sampled at an interval .

4. Mapping Formula in Impulse Invariance Method

The impulse invariance method maps the poles of the analog filter in the **s-plane** to the **z-plane**.

The mapping relation is:

where:

- = analogue complex frequency variable
- = digital complex frequency variable
- = sampling period

This exponential mapping ensures that the sampled impulse response of the analog filter becomes the impulse response of the digital filter.

5. Significance of Impulse Invariance Method

The impulse invariance technique has several advantages:

- **Preserves time-domain characteristics** of the analogue filter.

- **Simple mathematical implementation**.
- **Direct relationship between analogue and digital impulse responses**.
- **Easy implementation in MATLAB** using built-in functions like impinvar.

Because of these benefits, it is often used when maintaining the **exact impulse response shape** is important.

6. Limitations of Impulse Invariance Method

Despite its advantages, the impulse invariance method has some limitations:

- **Aliasing in the frequency domain:** High-frequency components of the analogue system may overlap after sampling.
- **Not suitable for high-pass filters:** Because aliasing distorts the frequency response.
- **Limited control over frequency response:** Compared to the bilinear transformation.
- **Requires sufficiently high sampling frequency** to reduce aliasing effects.

Due to these limitations, other methods like **bilinear transformation** are often preferred in many digital filter designs.

7. Applications of IIR Filters

IIR filters are used in many practical systems, including:

- Audio signal processing
- Speech processing
- Noise reduction systems
- Biomedical signal processing (ECG, EEG filtering)
- Communication systems

- Radar and sonar signal processing
- Image processing
- Control systems

Their efficiency and low computational cost make them ideal for real-time DSP applications.

MATLAB Commands used:

PROGRAM

```
% DSP Experiment - Analog to Digital Transfer Function using Impulse Invariance

clc;
clear;

% First Analog Transfer Function
a = [0 0 1];
b = [1 3 2];

H1_s = tf(a,b)
[num1, den1] = impinvar(a,b,1);
H1_z = tf(num1,den1,1)

% Second Analog Transfer Function
c = [1 0.1];
d = [1 0.2 9.01];

H2_s = tf(c,d)
[num2, den2] = impinvar(a,b,0.1);
H2_z = tf(num2,den2,0.1)

% Third Analog Transfer Function
e = [2.8 4.9 2.9];
```

f = [1 3 2.85 1.7];

H3_s = tf(e,f)
[num3, den3] = impinvar(a,b,0.1);
H3_z = tf(num3,den3,0.1)
RESULTS

The first picture shows the output of the first inputs(number of poles), and the output contains the equation with constant time:

```
H1_z =

         0.2325 z
  ------------------------
  z^2 - 0.5032 z + 0.04979

Sample time: 1 seconds
Discrete-time transfer function.
```

The second picture shows the output of the first input (number of poles), and the output contains the equation with constant time:

```
H2_z =

        z - 0.9048
  ---------------------
  z^2 - 1.81 z + 0.8187

Sample time: 1 seconds
Discrete-time transfer function.
```

The third picture shows the output of the first input (number of poles), and the output contains the equation with constant time:

```
H3_z =

       2.8 z^2 - 1.65 z + 0.2575
  -------------------------------------
  z^3 - 0.553 z^2 + 0.07484 z - 0.002479

Sample time: 1 seconds
Discrete-time transfer function.
```

CONCLUSION

1. **When Sampling Time is Constant**

- The mapping from analogue to digital domain remains fixed.
- The relation stays constant.
- The digital transfer function does not change with time.
- The system behaves as a Linear Time-Invariant (LTI) system.
- Filter coefficients remain constant.
- The frequency response is stable and predictable.
- The digital filter accurately represents the sampled analog impulse response.

2. **When Sampling Time Varies**

- The mapping between -plane and -plane changes continuously.
- The relation changes with time.
- The digital transfer function becomes time-dependent.
- The system behaves as a time-varying system.
- Filter coefficients change over time.
- The frequency response becomes inconsistent.
- The filter may produce distorted or unstable output.